Command Broker Implementation Guide

Governing the Human Decision Authority Requirement

in AI-Augmented Industrial Control Systems

Implementing the Bright Line Criteria of the
ISA Tiered Assessment Framework for Cognitive Errors

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Intended Audience: Asset Owner/Operators of Critical Infrastructure

Parent Documents: ISA Parent Framework V1.2 | IEEE V&V Cognitive AI Framework V1.3

1 Purpose and Scope

This guide is addressed to the people who own and operate critical infrastructure facilities deploying generative AI in industrial control environments. It is not addressed to standards committees, although it is written to be consistent with the parent frameworks that those committees are developing. Its purpose is practical: to explain what the Command Broker function is, what it demands of the person performing it, how an Asset Owner/Operator (AOO) discharges their duty of care in establishing and sustaining that function, and how to train, test, and credential the individuals on whom the entire safety architecture depends.

The ISA Parent Framework V1.2 and the IEEE V&V Cognitive AI Framework V1.3 establish the Command Broker as an architectural requirement. This human decision authority must stand between AI-generated recommendations and any control action affecting critical infrastructure above the Bright Line. Those documents are addressed primarily to standards developers and AI system suppliers. They define what the architecture must prevent. This guide addresses the other half of the problem: what the architecture demands of the humans it depends on.

A recurring theme in this guide is that the Command Broker is not a job title, a checkbox, or a warm body in the control room. It is a defined function with specific competency requirements, specific cognitive demands, and specific accountability obligations. An AOO who deploys a generative AI system above the Bright Line and places an unqualified individual in the Command Broker role has not satisfied the architectural requirement. They have created the appearance of compliance while leaving the protective function hollow — precisely the failure mode this entire framework is designed to prevent.

This guide applies to all critical infrastructure sectors deploying generative AI in Tier 1 (safety-critical) or Tier 2 (quality-of-service) applications as defined in the ISA Parent Framework. Sector-specific requirements — the particular regulatory training mandates in the nuclear, electric, water, and other sectors — are addressed in the relevant sector application guides. The duty of care obligations and Command Broker qualification framework presented here are sector-neutral and apply regardless of whether a sector regulator has yet addressed AI-augmented operations in its standards.

2 What the Bright Line Requires and Why

The ISA Parent Framework's Bright Line Criteria prohibit fully autonomous generative AI operation in any application where a cognitive error could produce safety-critical or significant quality-of-service consequences. Above the Bright Line, the AI system may analyze, synthesize, and recommend — but it may not command. Every action affecting the controlled process must pass through a human decision-maker who evaluates the AI's recommendation and authorizes or rejects it. That human is the Command Broker.

The rationale for this requirement is not a commentary on the capability or quality of any particular AI system. It reflects a fundamental property of the technology. Generative AI systems based on large language models are non-deterministic: the same input may produce different outputs, the space of possible responses cannot be exhaustively enumerated, and there will always be conditions that were not anticipated during qualification testing. No matter how thorough the cognitive error assessment process is, it cannot eliminate the possibility of an error that testing did not detect. In high-consequence applications, that residual possibility is unacceptable without a human safeguard whose judgment is independent of the AI system's assessed performance.

The analogy from the IEEE V&V Cognitive AI Framework is useful here. The Command Broker's protective function is architectural, not procedural. It does not depend on the operator's ability to detect every AI error independently. It depends on the physical separation between AI recommendation and control action — the fact that the AI system cannot directly command changes to setpoints, valve positions, breaker states, or other process control parameters without affirmative human authorization. That separation is what makes the architecture sound. But the architecture's soundness does not guarantee the function's effectiveness. A correctly implemented Command Broker interface that routes AI recommendations through a human who rubber-stamps every output is architecturally compliant and functionally worthless.

This is the gap this guide addresses. The standards can require the interface. They can prohibit the AI from having a direct pathway to the control system. What they cannot guarantee — without AOO-level governance obligations — is that the human positioned in the Command Broker role is actually exercising the judgment the architecture assumes.

3 The Command Broker: What It Looks and Feels Like

Before addressing qualifications and training, it is worth establishing a concrete picture of what the Command Broker function actually involves in practice. Standards language necessarily abstracts; AOOs need to understand the operational reality their people will inhabit.

3.1 The Operational Environment

The Command Broker sits at the intersection of two information streams that were previously separate: the conventional human-machine interface (HMI) that has always been present in the control room, and the AI advisory interface that presents generated analysis, pattern recognition, and recommendations. The AI interface is conversational in character — it can respond to natural language questions, explain its reasoning, and present alternative scenarios. It is also fast, confident in tone, and frequently correct. These properties are precisely what make it valuable and precisely what make it dangerous if the Broker's critical faculties are not engaged.

In a typical Tier 1 deployment, the AI might monitor sensor data continuously and surface an alert: it has identified a pattern consistent with developing equipment stress and recommends an operational adjustment. The recommendation arrives with supporting data — the sensor readings, the historical pattern it matched, the confidence level, and the potential consequence if the condition progresses. The Broker's job is not simply to read the recommendation and execute it. The Broker must evaluate whether the AI's pattern recognition is plausible given current plant conditions, whether the supporting data is internally consistent, whether there are factors the AI cannot see or may be misweighting, and whether the recommended action is appropriate given the full operational context. Only then does the Broker authorize the adjustment through the conventional control interface.

This evaluation is not a formality. It is work. It requires the Broker to know the process well enough to recognize when the AI is right, when the AI is plausibly right, but the action carries risks not reflected in the recommendation, and when the AI is confidently wrong. That last case — confident wrongness — is the most dangerous and the one least addressed by conventional operator training. Operators trained before AI deployment have learned to treat instrumentation as a truth-telling system. A pressure sensor reads 47 PSI; that is what the pressure is. A generative AI system that states a recommended setpoint with the same grammatical confidence as a sensor reading is doing something categorically different: it is generating a plausible output given its training and the current context, which is not the same thing as measuring a physical quantity. Closing the gap between how AI output feels and what it actually represents is one of the core cognitive challenges the Command Broker training program must address.

3.2 The Texture of the Decision

The Command Broker's decision is not binary — authorize or reject. In practice, the decision space has several dimensions that qualified Brokers must navigate.

The first dimension is verification of the AI's factual basis. The AI's recommendation rests on data it has observed and patterns it has recognized. The Broker should be able to verify the key data points against the conventional HMI independently. If the AI's recommendation is premised on a sensor reading or process condition that the Broker cannot confirm through conventional instrumentation, that is a flag — not necessarily a disqualifying one, but a reason to probe further before authorizing.

The second dimension is consistency with the Broker's operational knowledge. An experienced process operator has mental models of how the facility behaves under normal conditions, under stress, and under the range of abnormal conditions the facility has encountered. The AI's recommendation should be evaluated against those mental models. A recommendation that is technically coherent in isolation but inconsistent with how the plant actually behaves is a candidate for rejection or escalation regardless of the AI's stated confidence.

The third dimension is consequence awareness. The Broker must understand not only what the AI is recommending but what will happen if that recommendation is executed — including what will happen if it turns out to be wrong. Some incorrect AI recommendations produce minor and easily corrected consequences. Others, if acted upon in the wrong operational context, could accelerate a developing problem rather than mitigate it. The Broker's consequence awareness must be sufficient to differentiate these cases.

The fourth dimension is time pressure. Many control room environments involve time-sensitive decisions. The AI may be surfacing a recommendation at a moment when the Broker is already managing competing demands on their attention. The training program must address how to handle AI recommendations under time pressure — specifically, the conditions under which it is appropriate to defer, escalate, or reject a recommendation when there is insufficient time for a full evaluation.

3.3 The Accountability Structure

The Command Broker is accountable to the AOO, not to the AI system. This is a point that sounds obvious but has practical implications that need to be made explicit in AOO governance documentation and operator training.

When the Broker authorizes an AI recommendation and executes the corresponding control action, the Broker owns that action. The AI does not make decisions; the Broker does. If the action produces an undesirable outcome, the accountability flows to the Broker and through the Broker to the AOO — not to the AI system, not to the AI supplier. This is not a punitive observation; it is a governance reality that must be understood by everyone involved. Brokers who understand that they own their decisions are more likely to exercise genuine evaluation rather than reflexive authorization. AOOs who understand that they own the Broker function are more likely to invest in qualifying and training the people performing it.

This accountability structure is also what connects the Command Broker requirement to the legal standards of duty of care and skill of the craft. These doctrines apply independently of regulatory requirements. A court evaluating a harmful incident at a facility operating an AI-augmented control system will ask whether the AOO exercised reasonable care in ensuring the humans performing safety-significant functions were qualified to perform them. The existence of a Command Broker interface does not answer that question. Evidence that the Broker was trained, tested, credentialed, and regularly exercised against AI-specific failure scenarios is what answers it.

4 Duty of Care and Skill of the Craft

Two legal standards define the AOO's obligations with respect to Command Broker qualification, independent of any regulatory requirement. Understanding these standards is not an optional background for legal staff; it is an essential context for the operational decisions AOOs make when deploying AI-augmented control systems.

4.1 Duty of Care

Duty of care is the obligation to exercise reasonable care to prevent foreseeable harm to others. In the context of critical infrastructure operations, this duty is well-established: utilities, water systems, industrial facilities, and their operators have long been held to a duty of care toward the public, their workers, and the environment. What changes with AI deployment is the scope of what reasonable care requires.

An AOO deploying a generative AI system in a Tier 1 or Tier 2 application, knowing that the Bright Line prohibits autonomous AI action and requires a human Command Broker, has constructive knowledge that the Command Broker function is safety-significant. Constructive knowledge of a safety-significant function carries an obligation to ensure the people performing that function are qualified to perform it. Deploying an unqualified Broker — or deploying the Command Broker interface without any process for determining who qualifies to use it and under what conditions — is a foreseeable failure mode that a reasonable AOO should have anticipated and prevented.

The duty of care argument does not require a regulator to have written a rule. It requires only that a court, or an insurer, or a counterparty in litigation, find that a reasonable AOO in the same position should have known that an unqualified Broker created foreseeable risk. In the current environment — where the ISA, IEEE, and other standards bodies are actively publishing guidance on Command Broker qualification requirements — that finding becomes progressively easier to sustain. Published standards become the benchmark against which reasonable care is measured.

4.2 Skill of the Craft

Skill of the craft is the competency standard that defines what a qualified practitioner in a given role is expected to know and be able to do. It is not a fixed regulatory threshold; it is a community-of-practice standard that evolves as technology and practice evolve. It applies in fields where practitioners are expected to exercise professional judgment, and it can be invoked as a standard of care in litigation even where no specific regulation mandates it.

For Command Broker operators, the skill of the craft is currently being defined. There is no mature community of practice for AI-augmented control room operations. This creates both a risk and an opportunity. The risk is that AOOs operating without guidance may establish practices that fall below what will eventually be recognized as the competency standard, exposing themselves to liability in retrospect. The opportunity is that AOOs and standards developers who establish sound practices now are participating in defining what skill of the craft means for this function, and published guidance from recognized standards bodies becomes part of that definition.

The qualification framework in Section 5 of this guide is intended to contribute to that definition. It identifies the knowledge domains, skills, and demonstrated capabilities that a qualified Command Broker should possess. As this framework matures through the standards process, these requirements will become part of the community's understanding of what a competent Broker looks like.

4.3 The Insurance Dimension

Insurers writing coverage for critical infrastructure facilities are increasingly attentive to AI deployment practices. A facility that can demonstrate a documented, auditable Command Broker qualification process — tied to specific competency requirements, tested through structured exercises, and maintained through continuing education — presents a materially different risk profile than one that cannot. The qualification framework is not merely a legal protection; it is an underwriting consideration that can affect coverage terms and premiums.

AOOs should anticipate that insurers will, as AI deployment in critical infrastructure matures, require evidence of Command Broker qualification as a condition of coverage for AI-related operational failures. Establishing the program now, before that requirement is formalized, places the AOO in a demonstrably better position both in underwriting and in the event of a claim.

5 Command Broker Qualification Requirements

The following requirements define what an individual must know and be able to demonstrate to be qualified as a Command Broker for Tier 1 or Tier 2 AI-augmented control applications. Requirements are organized into three domains: process domain competency, AI-interaction competency, and decision authority competency. All three domains are required; competency in one does not substitute for a deficiency in another.

Tier 1 applications require demonstrated competency at a higher level of rigor across all three domains than Tier 2 applications. Section 5.4 identifies the specific differentiation.

5.1 Process Domain Competency

The Command Broker must possess operational expertise in the specific process being managed. This is the baseline requirement and the one most easily satisfied through existing qualification frameworks, since regulated sectors generally have established operator certification or qualification processes for control room personnel.

Process domain competency for Command Broker purposes includes: knowledge of the facility's normal operating envelope, including the setpoint ranges, process limits, and alarm conditions that define acceptable operation; familiarity with the facility's abnormal and emergency operating procedures, including the conditions that trigger each and the expected system response; understanding of the causal relationships between process variables sufficient to evaluate whether an AI recommendation is consistent with expected plant behavior; and knowledge of the specific failure modes and degradation patterns relevant to the equipment and processes the AI system is advising on.

An existing regulatory certification — an NRC reactor operator license, a NERC system operator certification, a state-issued water treatment operator certificate — satisfies the baseline process domain competency requirement for the scope of that certification. AOOs must verify that the certification scope covers the specific process and application for which the individual will serve as Command Broker.

5.2 AI-Interaction Competency

AI-interaction competency is the domain that existing operator qualification frameworks do not address, and that this guide is most specifically designed to establish. A qualified Command Broker must understand how generative AI systems behave, how they fail, and how to recognize the difference between a trustworthy recommendation and a plausible-sounding error.

The core knowledge requirements in this domain are as follows.

First, the Broker must understand the difference between deterministic and non-deterministic systems. Conventional instrumentation and control systems are deterministic: given the same inputs, they produce the same outputs, and their behavior can be formally verified. Generative AI systems are non-deterministic: the same input may produce different outputs, and the space of possible responses cannot be exhaustively tested. This difference has practical implications for how much weight to place on AI output and under what conditions to require independent verification.

Second, the Broker must understand AI hallucination as a failure mode — specifically, that a generative AI system can produce confident, grammatically coherent, technically plausible output that is factually wrong. The mechanism is not the same as a sensor malfunction; the system is not producing a known error code or a clearly anomalous reading. It is producing a response that, by the standards it was trained on, looks correct. The ability to recognize hallucination risk in high-stakes recommendations — and to know when to require external verification before acting — is a core competency for Command Broker qualification.

Third, the Broker must understand the concept of the AI system's operational domain — the conditions under which it has been trained and for which its performance has been characterized. Recommendations that fall at or near the boundaries of that domain, or that involve conditions the system has not been tested on, carry elevated uncertainty that the Broker must account for. The AI system's provenance documentation, required by the ISA Parent Framework, provides the information the Broker needs to make this assessment; the Broker must know how to read and apply it.

Fourth, the Broker must be familiar with automation complacency as a cognitive failure mode — the tendency for operators who regularly see AI recommendations confirmed to reduce the rigor of their independent evaluation progressively. Complacency is not a character flaw; it is a documented human factors phenomenon that training must specifically address and that exercise programs must specifically probe.

5.3 Decision Authority Competency

Decision authority competency addresses the Broker's capacity to exercise independent judgment and to maintain that independence under operational pressure. This domain is the hardest to assess through conventional examination but is arguably the most important: the architecture fails precisely when the Broker defers to the AI instead of exercising independent evaluation.

Decision authority competency requires that the Broker understand they own their decisions and that AI recommendations do not transfer accountability. It requires demonstrated ability to reject or modify AI recommendations when independent evaluation indicates the recommendation is wrong, incomplete, or contextually inappropriate. It requires the ability to articulate the basis for authorization or rejection — not at the level of filing a written justification for every routine action, but at the level of being able to explain, if asked, why a recommendation was acted upon or why it was not. And it requires familiarity with the escalation pathways available when the Broker is uncertain — when to involve a supervisor, when to request a second evaluation, and when to take no action pending resolution of the uncertainty.

5.4 Tier Differentiation

Competency Domain	Tier 1 (Safety-Critical)	Tier 2 (Quality-of-Service)
Process Domain	Current regulatory certification required (NRC, NERC, state, or equivalent). The scope must cover the specific application. Annual refresher required.	Demonstrated operational experience in the process domain. Certification required where available—biennial refresher.
AI-Interaction	Formal instruction covering all four AI-interaction competency areas. Minimum 16 hours of initial training. Annual refresher with updated content as AI systems change.	Formal instruction covering all four areas. Minimum 8 hours of initial training. Biennial refresher.
Decision Authority	Structured scenario-based assessment under adversarial conditions (AI deliberately presenting incorrect	Structured scenario-based assessment. Tabletop format acceptable for initial qualification. Biennial reassessment.

Competency Domain	Tier 1 (Safety-Critical)	Tier 2 (Quality-of-Service)
	recommendations). Assessment must be passed before the operational assignment—annual reassessment.	
Documentation	Individual qualification record maintained by AOO. Training, assessment, and exercise records are retained for the facility operating life plus 10 years.	Individual qualification record maintained. Records retained for 7 years.

6 Lesson Plans and Learning Objectives

The following section converts the competency requirements in Section 5 into structured instructional content. Each module is defined by its learning objectives, instructional approach, and assessment method. The modules are designed to be delivered as a coherent program; they may also be used modularly where an individual's existing qualification satisfies some domain requirements.

Module A: Foundations of AI Behavior in Control Environments

Target audience: All Command Broker candidates. Tier 1 and Tier 2.

Estimated duration: 4 hours (Tier 1) / 3 hours (Tier 2).

Delivery: Formal instruction, case study discussion.

Learning Objectives: Upon completion, the candidate will be able to distinguish generative AI from deterministic control system components and explain the practical implications of non-determinism for operational reliance; describe the AI hallucination failure mode in terms applicable to control system advisory functions and identify the conditions under which hallucination risk is elevated; explain what an AI system's operational domain means and how to apply provenance documentation to assess whether a recommendation falls within or outside the characterized domain; and explain automation complacency as a cognitive failure mode and describe the behavioral indicators that suggest complacency is developing in an operator or team.

Instructional approach: The module opens with the air-gap parallel — the observation that 'there is a human in the loop' functions as reassurance language in the same way that 'the system is air-gapped' did, and that both phrases require examination rather than acceptance. Case studies drawn from aviation, nuclear, and process industries illustrate automation complacency incidents; facilitators draw explicit parallels to the AI advisory context. The module includes hands-on exposure to a representative AI advisory interface so candidates experience the texture of AI-generated recommendations before they encounter them operationally.

Module B: Evaluating AI Recommendations — The Broker's Analytical Framework

Target audience: All Command Broker candidates. Tier 1 and Tier 2.

Estimated duration: 4 hours (Tier 1) / 3 hours (Tier 2).

Delivery: Formal instruction with worked examples from the candidate's process domain.

Learning Objectives: Upon completion, the candidate will be able to apply a structured four-part evaluation framework to any AI recommendation — verifying the factual basis, checking consistency with process knowledge, assessing consequence implications, and managing time pressure appropriately; identify the specific indicators that should trigger independent verification before authorizing a recommendation; distinguish between AI recommendations that warrant immediate authorization, recommendations that warrant modification before authorization, recommendations that warrant deferral pending further evaluation, and recommendations that should be rejected outright; and demonstrate the ability to probe an AI system's reasoning through natural language interaction to expose the basis for a recommendation.

Instructional approach: The module is built around worked examples using process-domain-specific scenarios. For each scenario, the candidate receives an AI recommendation and is walked through the four-part evaluation framework. Multiple scenarios are presented where the AI recommendation is correct; a subset presents recommendations that are subtly or significantly wrong. The emphasis is on developing the habit of evaluation rather than the habit of acceptance. Candidates are required to verbalize their reasoning for each authorization or rejection decision, establishing the expectation that decision authority involves articulable judgment.

Module C: Accountability, Escalation, and the Regulatory Context

Target audience: All Command Broker candidates. Tier 1 and Tier 2.

Estimated duration: 2 hours.

Delivery: Formal instruction, scenario discussion.

Learning Objectives: Upon completion, the candidate will be able to explain the Command Broker accountability structure — that authorization of an AI recommendation is the Broker's decision and not the AI system's; describe the facility's escalation pathway for AI recommendations the Broker is uncertain about, including the criteria for involving a supervisor or requesting a second evaluation; identify the circumstances under which taking no action pending resolution of uncertainty is the appropriate Command Broker decision; and describe the facility's documentation requirements for Command Broker decisions in the operational log or equivalent record.

Instructional approach: This module addresses the cultural and institutional dimensions of Command Broker operation that are not addressed by technical training. It includes discussion of the difference between healthy skepticism of AI output and unhealthy over-reliance, and the organizational factors — supervisor pressure, production demands, staffing levels — that can erode Command Broker independence. Facilitators should be prepared to discuss real organizational pressures that candidates may face; this module works best as a facilitated discussion rather than a lecture.

Module D: Sector-Specific Regulatory and Compliance Context

Target audience: All Command Broker candidates. Content varies by sector.

Estimated duration: 2 hours (Tier 2) / 4 hours (Tier 1).

Delivery: Formal instruction by a sector-knowledgeable instructor.

Learning Objectives: Upon completion, the candidate will be able to identify the specific regulatory requirements governing AI-augmented operations at the facility, including any applicable NERC, NRC, EPA, PHMSA, or state-level requirements; explain how the Command Broker function satisfies the facility's Bright Line Criteria compliance obligation; and describe the facility's AI system provenance documentation and explain how to use it to assess recommendation reliability.

Instructional approach: Module content is facility- and sector-specific and must be developed by the AOO in consultation with their regulatory counsel and compliance staff. This module is listed last in the sequence because it presupposes the foundational understanding developed in Modules A through C.

7 Tabletop Exercise: Command Broker Decision Under Pressure

7.1 Exercise Purpose and Design Principles

Tabletop exercises serve two functions in the Command Broker qualification program. For initial qualification, they provide a structured assessment environment in which candidates must demonstrate decision authority competency under conditions that simulate the time pressure, competing information, and AI-generated outputs they will encounter operationally. For continuing qualification, periodic exercises probe for complacency development and test whether Brokers maintain their evaluative discipline over time.

The fundamental design principle for Command Broker exercises is adversarial realism: at least one-third of AI recommendations presented during the exercise must be wrong in ways the candidate can detect through careful evaluation but might miss under time pressure or complacency. If every AI recommendation in an exercise is correct, the exercise trains authorization behavior rather than evaluation behavior. The goal is to calibrate the candidate's ability to catch the AI being wrong — not to test whether they can authorize the AI when it is right.

Exercises should be designed with the candidate's process domain in mind. Abstract scenarios are less effective than scenarios drawn from the specific facility type, equipment, and operational conditions the candidate will encounter. AOOs with limited internal exercise design capability should consider engaging sector peers, industry associations, or subject matter experts to develop scenario libraries appropriate to their operations.

7.2 Exercise Structure

The following exercise structure is designed for a three-hour tabletop session for Tier 1 qualification. A two-hour version suitable for Tier 2 qualification is noted in parentheses where the structure differs.

Phase 1 — Orientation (20 minutes): The facilitator briefs participants on the exercise scenario — the facility, the operational context, and the AI system that is deployed. Participants receive the AI system's operational profile summary (a brief document summarizing the system's process domain, training scope, and known limitations) and are given time to review it. This phase establishes that the exercise is not a test of process knowledge alone but specifically of Command Broker judgment in an AI-augmented environment.

Phase 2 — Baseline Operations (30 minutes, Tier 2: 20 minutes): The AI advisory system is operating normally and generating accurate recommendations. This phase establishes a baseline of correct AI performance that candidates will experience before encountering deviations. The purpose is to calibrate the complacency dynamic: candidates who have authorized several correct recommendations without incident are more vulnerable to accepting an incorrect one.

Phase 3 — Anomaly Injection (60 minutes, Tier 2: 40 minutes): The facilitator introduces a developing operational condition that the AI system has not been trained on or is misinterpreting. The AI continues to generate recommendations, but they are now subtly wrong — plausible in isolation but inconsistent with the developing situation when evaluated carefully. Candidates must identify the discrepancy, probe the AI's reasoning, verify against conventional instrumentation, and make a Command Broker decision. The facilitator presents the AI output verbally or on a screen; candidates respond as they would operationally. Observers note whether candidates verbalize their evaluation reasoning or authorize without articulating their basis.

Phase 4 — Escalating Pressure (30 minutes, Tier 2: 20 minutes): Time pressure and competing demands are introduced. A secondary alarm condition requires the Broker's attention simultaneously with an AI recommendation. The AI's recommendation in this phase is designed to appear more urgent than the alarm condition, inverting the actual priority order. This phase assesses whether candidates maintain structured evaluation under pressure or default to expedient authorization.

Phase 5 — Debrief (40 minutes, Tier 2: 30 minutes): Facilitator-led debrief reviewing each Command Broker decision point. For each AI recommendation encountered during the exercise, the debrief addresses: what the AI recommended and why, what information was available to evaluate the recommendation, what the correct decision was and why, and what the candidate actually decided and what drove that decision. The debrief is not punitive; its purpose is to make visible the cognitive process that good Command Broker judgment requires.

7.3 Qualification Assessment Criteria

Candidates qualify when they demonstrate the following across the full exercise: independent verification of AI recommendations against conventional instrumentation at least once during Phase 3; correct identification of at least two-thirds of the incorrect AI recommendations presented; at least one instance of recommendation rejection or modification with articulable basis; appropriate escalation or deferral during Phase 4 rather than expedient authorization; and post-exercise ability to articulate the reasoning behind at least three Command Broker decisions made during the exercise.

Candidates who do not meet these criteria on the initial attempt may repeat the exercise after additional instruction in the domain where the deficiency was observed. A candidate who fails to identify any incorrect AI recommendations — demonstrating authorization behavior regardless of recommendation quality — should not be qualified for Tier 1 deployment and should receive targeted training on AI hallucination and adversarial scenario recognition before reassessment.

7.4 Sample Scenario: Water Treatment Facility, Tier 1

The following sample scenario illustrates the exercise design principles in a drinking water treatment context. AOOs in other sectors should adapt the scenario to their process domain.

Scenario Setup: A municipal drinking water treatment facility operates an AI advisory system for chemical dosing and process optimization. The system has been in operation for six months and has a good track record on routine dosing recommendations. The Broker on shift today is an experienced operator with a current state certification. It is a normal Wednesday afternoon.

Phase 2 Baseline: The AI recommends a minor adjustment to chlorine dosing based on incoming turbidity data — consistent with recent practice and confirmed by conventional instrumentation. The Broker authorizes. The AI recommends holding current pH adjustment setpoints despite a slight influent pH shift — the shift is within the AI's trained normal range. The Broker authorizes. Two routine, correct recommendations establish a baseline.

Phase 3 Anomaly: Upstream, a runoff event following a heavy rain has introduced an unusual organic loading to the raw water source. The turbidity sensors are reading within normal range because the organic material is dissolved rather than particulate. The AI, trained on turbidity as the primary indicator of influent quality variation, does not recognize the dissolved organic loading and recommends reducing chloramine contact time based on its turbidity-based model — a recommendation that, if followed, would result in inadequate disinfection and a potential public health violation. The conventional water quality analyzer, which the AI does not have access to, shows an elevated TOC reading that the Broker can see on the conventional HMI. The Broker's task is to notice the discrepancy, probe the AI's reasoning, recognize that the AI's recommendation is premised on incomplete information, and reject the recommendation pending a process engineer review.

Phase 4 Pressure: While managing the Phase 3 anomaly, a routine pump alarm activates on a secondary system. The AI recommends prioritizing the pump issue and proceeding with the reduced contact time to maintain throughput. This recommendation inverts the actual priority: the potential public health consequence of inadequate disinfection significantly outweighs the pump alarm. The Broker must maintain priority discipline under the competing demands of two simultaneous issues and an AI recommendation that appears to support expedient resolution of both.

Debrief focus areas: recognition of the turbidity-TOC discrepancy as an indicator of AI operational domain exceedance; probe questions the Broker could have asked the AI to expose the gap in its reasoning; escalation pathway — when should the process engineer have been called, and what information should have been provided; and priority judgment in Phase 4.

7.5 Sample Scenario: Balancing Authority Control Room, Tier 1

The following scenario is set in an electric utility Balancing Authority control room operating under NERC CIP-002 High Impact BES Cyber System obligations. The AI advisory system has been deployed to assist the system operator with real-time operational decisions affecting the reliability of the bulk electric system. This scenario illustrates a failure mode specific to the electric sector: an AI system that is performing correctly within its training domain but whose recommendation, if executed under the developing conditions, could contribute to cascading consequences that the AI is not positioned to anticipate.

Scenario Setup: A regional Balancing Authority operates an AI advisory system that monitors real-time system frequency, interchange schedules, and generation dispatch data. The system provides the on-duty system operator with recommendations for corrective actions when the area control error (ACE) deviates beyond acceptable bounds or when the operator requests operational analysis. The AI system has been in operation for four months and has demonstrated reliable performance on routine ACE correction and economic dispatch advisory functions. The Command Broker on shift is a NERC-certified system operator with seven years of experience, currently in the last two hours of a twelve-hour shift on a mid-week afternoon.

Phase 2 Baseline: Over the preceding ninety minutes, the AI has made three advisory recommendations: a minor dispatch adjustment to a peaking unit to bring ACE within bounds following an unplanned generator trip in an adjacent Balancing Authority, a schedule tag correction identifying a clerical error in an interchange transaction, and an alert flagging a thermal limit approach on a major transmission path with a recommendation to reduce flow through economic redispatch. All three recommendations were correct, verified against SCADA data by the operator, and authorized. The operator has developed confidence in the system through this shift and the preceding weeks of operation.

Phase 3 Anomaly: An unplanned outage on a major 345 kV transmission line has just occurred in an adjacent Balancing Authority. The event is recent enough that the operator has received only the initial notification; full situational awareness of the post-contingency system state is still developing. The AI advisory system, analyzing the operator's area ACE and the visible generation and load data, recommends a significant upward dispatch adjustment to a large thermal unit within the operator's Balancing Authority to arrest a frequency deviation. The recommendation is consistent with the AI's training on ACE correction scenarios and is presented with high confidence.

What the AI cannot see: the transmission line outage has changed the flow patterns on the interconnected system in a way that makes the recommended dispatch potentially problematic. The thermal unit the AI is recommending for dispatch increases feed power through a path that, given the changed network topology, would push flow on a second transmission element toward its emergency rating. The AI's training did not include the specific post-contingency topology now present, and its model of the transmission system reflects the pre-outage network. The SCADA system on the operator's conventional console is beginning to show the changed flows, but the data is still updating, and the operator has not yet processed it. The AI's recommendation, which would have been correct thirty minutes ago, is potentially wrong now in a way that could require emergency action to reverse.

The Broker's task: recognize that the contingency event has changed the operating context in a way that may have invalidated the AI's recommendation; verify the developing transmission flow picture against SCADA before authorizing the dispatch increase; contact the adjacent Balancing Authority to confirm the post-contingency system state; and either defer the AI's recommendation pending verification, authorize a reduced dispatch adjustment within known safe limits, or reject the recommendation outright until situational awareness is established. The key evaluative question is whether the operator recognizes that an AI recommendation correct for pre-contingency conditions may not be correct for a post-contingency network that the AI has not yet processed.

Phase 4 Pressure: While the operator is evaluating the dispatch recommendation, the ACE continues to deviate, and the AI escalates its advisory urgency. A second notification arrives from the reliability coordinator, indicating that the interconnection is experiencing frequency deviation and requesting that Balancing Authorities with available capacity respond. The AI now displays both its original dispatch recommendation and an updated recommendation at a higher dispatch level, framing the urgency in terms of interconnection frequency support obligations. The operator is now managing three concurrent demands: the developing transmission flow concern on SCADA, the AI's escalating dispatch recommendation, and the reliability coordinator communication. The AI's framing of the situation as an interconnection frequency emergency is correct in its general characterization. Still, it does not account for the local transmission constraint that may make a large dispatch increase the wrong response.

Debrief focus areas: recognition that a contingency event constitutes a change in operating context that should trigger re-evaluation of any pending AI recommendations rather than continued reliance on pre-contingency analysis; the specific SCADA data elements the operator should have checked before authorizing the dispatch recommendation and at what point in the sequence; the reliability coordinator communication as a source of situational awareness independent of the AI system and how to integrate it into the Command Broker evaluation; the appropriate response to AI recommendations presented under urgency framing—whether the urgency of the frequency event changes the obligation to verify, or whether it makes verification more important rather than less; and the escalation pathway available to the operator when situational awareness is insufficient to evaluate a time-sensitive recommendation with confidence.

Electric sector design note: This scenario is deliberately set during the developing phase of a contingency event rather than during steady-state operations, because the developing contingency is where the AI's operational domain boundaries are most likely to be encountered in practice. An AI system trained on historical operational data will have seen many contingency events, but will not have been trained on every possible post-contingency network topology. The scenario probes the specific Command Broker competency most critical in the electric sector: recognizing when real-time system state has diverged from the state the AI's recommendation assumes, and holding that recognition against the pressure of urgency framing and recent AI performance that was entirely correct.

Facilitators designing electric sector exercises should consult with reliability coordinators and operating experience programs when developing post-contingency topology scenarios, as the specific transmission paths and thermal limits will vary significantly by region. The scenario structure above is transportable; the network specifics must be localized.

8 AOO Governance Obligations

The qualification framework in this guide imposes obligations on the AOO as an institution, not only on individual Command Broker candidates. This section summarizes those obligations in terms suitable for incorporation into AOO governance documentation, operational procedures, and management system requirements.

The AOO shall maintain a written Command Broker qualification program that identifies the competency requirements applicable to each AI-augmented application in the facility, the training and assessment processes through which candidates demonstrate those competencies, the credentialing process by which qualified individuals are authorized to perform the Command Broker function, and the ongoing exercise and recertification requirements for maintaining qualification.

The AOO shall maintain individual qualification records for each person authorized to perform the Command Broker function. These records shall document initial training and assessment results, exercise participation and performance, continuing education, and any incidents or observations that bear on the individual's Command Broker performance. Records shall be maintained for the periods specified in Section 5.4.

The AOO shall conduct a periodic review of the Command Broker qualification program, not less than annually for Tier 1 applications, to assess whether the program remains current with changes to the deployed AI systems, changes in operational conditions, and developments in the broader understanding of AI-related human factors. Updates to AI system provenance — including model updates, fine-tuning changes, or changes in retrieval augmentation sources — trigger a mandatory review of whether existing Broker qualifications remain adequate for the updated system.

The AOO shall establish and document the escalation pathway available to Command Brokers who are uncertain about an AI recommendation, including the identification of supervisory personnel who can be engaged, the criteria for deferring action pending resolution of uncertainty, and the process for logging Command Broker decisions and the basis for those decisions.

The AOO bears a duty of care with respect to the qualification of personnel performing the Command Broker function that is independent of sector-specific regulatory requirements. This duty exists because the Command Broker function is safety-significant. After all, the AOO has constructive knowledge of that significance through the Bright Line Criteria and associated standards guidance, and because the harm that could result from an unqualified Broker performing the function is foreseeable. Compliance with the regulatory minimum, where a regulatory minimum exists, is necessary but not sufficient to discharge this duty. Where no regulatory minimum exists, the duty of care standard requires the AOO to establish qualifications consistent with the skill of the craft as defined by applicable standards guidance, including this document.

9 Document Control and Revision History

Version	Date	Author	Description
V0.1	June 2026	T. Roxey	Initial draft. Establishes Command Broker qualification framework, AOO governance obligations, lesson plans, learning objectives, and tabletop exercise structure for internal review.

10 Informative References

ISA Parent Framework for Cognitive Errors in Generative AI Systems Deployed in Industrial Control Environments, V1.2, Eclectic Technologies, April 2026.

IEEE Framework for Verification and Validation of Cognitive Behaviors in Generative AI Systems Deployed in Industrial Control Environments, V1.3, Eclectic Technologies, 2026.

Roxey, T.E. Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers. Version 29. Eclectic Technologies, 2026.

IEC 62443 Series, Security for Industrial Automation and Control Systems, International Electrotechnical Commission.

IEC 61508, Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems, International Electrotechnical Commission.

NIST AI Risk Management Framework (AI RMF 1.0), National Institute of Standards and Technology, January 2023.

NIST SP 800-82 Rev. 3, Guide to Operational Technology (OT) Security, National Institute of Standards and Technology.

Restatement (Third) of Torts: Physical and Emotional Harm, American Law Institute, 2010. (Duty of care doctrine.)